2. Python SQLite Game Shop Tutorial - Create Table

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Time required: 60 minutes

- Follow all directions carefully and accurately.
- Think of the directions as minimum requirements.

SQL Tutorial

- https://www.w3schools.com/sql/sql_create_table.asp

SQLite with Python Tutorials

- SQLite Databases with Python - Full Course – FreeCodeCamp.org
- https://www.sqlitetutorial.net

Cursor Object

Let's discuss the cursor object.

- The cursor object is used to make the connection to the database for executing text-based SQL queries.

- It acts as middleware or controller between the SQLite database connection and the SQL query. It is created after creating a connection to the SQLite database.
- The cursor is a control structure used to traverse the records of the database.
- All SQL commands are be executed using the cursor object only.

SQLite DataTypes

- **NULL:** The value is a NULL value.
- **INTEGER:** Store a whole number.
- **REAL:** Floating-point value, for example, 3.14, the value of PI.
- **TEXT:** A text string. TEXT value stored using UTF-8, UTF-16BE or UTF-16LE encoding.
- **BLOB:** The value is a blob of data, i.e., binary data. It is used to store images and files.

The following Python types convert to SQLite types.

Python Types	SQLite Types
None	NULL
int	INTEGER
float	REAL
str	TEXT
bytes	BLOB

Tutorial 1: Creating Tables

In an SQL database, data is stored in tables. Tables define a set of columns and fields, much like a spreadsheet. They contain 0 or more rows with data for each of the defined fields.

Let's create a table named **products** that tracks the following data. This is a data dictionary.

prod_id (primary key)	INTEGER
prod_name	TEXT
prod_price	REAL

prod_qty	INTEGER

We will create a **products** table using SQLite3 in Python.

SQL is a scripting language like Python. You can assign SQL code to a String variable using a multiline string.

The result of the following code is the same as what we did earlier in DB Browser for SQLite.

Do: sqlite_tutorial_2_create_table.py

Copy your previous program and save it as **sqlite_tutorial_2_create_table.py**

```python
"""
    Name: sqlite_tutorial_2_create_table.py
    Author: William Loring
    Created: 01/05/22
    Use SQLite with Python
"""
# Import sqlite3 database library
import sqlite3


class GameShop():

    """
    Represents a game shop management system.

    Attributes:
        database (str): The name of the database file.
    """

    def __init__(self, database: str):
        """
        Initializes the GameShop class.

        Parameters:
            database (str): The name of the database file.
        """

        # Assign the provided database name to the class attribute
        self.database = database
```

```python
# ------------------------- CREATE TABLE ----------------------------------#
    def create_table(self):
        """Create database and table if not exists."""
        # Create a connection object which will connect to or
        # create an SQLite database file
        connection = sqlite3.connect(self.database)

        # Create a cursor object to interact with the database
        cursor = connection.cursor()
        print("-- Connected to the database --")

        # SQL statement to drop 'products' table if it exists
        # SQL statement to create the 'products' table
        SQL = """
        DROP TABLE IF EXISTS products;
        CREATE TABLE products (
            prod_id         INTEGER PRIMARY KEY,
            prod_name       TEXT,
            prod_price      REAL,
            prod_qty        INTEGER
        );
        """

        # Execute the SQL script
        cursor.executescript(SQL)
        print(" Table created in memory")

        # Commit the changes to the database to write them to disk
        connection.commit()
        print(" Table committed to disk")

        # Close the cursor and the connection to prevent resource leaks
        cursor.close()
        connection.close()
        print("-- Connection closed --\n")


# Create a GameShop instance with the specified database file
game_shop = GameShop("game_shop_2_desc.db")

# Create the 'products' table in the database
game_shop.create_table()
```

Example run:

NOTE: Before you run this program, delete **game_shop_2.db** if it exists from a previous run.

```
-- Connected to the database --
   Table created
-- Connection closed --
```

Explanation

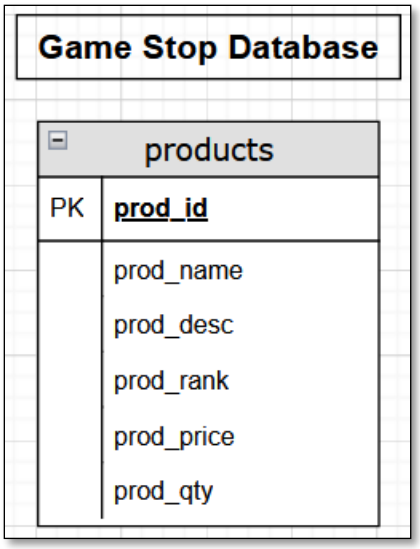
- The "**CREATE TABLE** "products" ..." multiline string is a SQL statement that creates a table named **products** with the columns described earlier:
    - **prod_id** of type INTEGER PRIMARY KEY
    - **prod_name** of type TEXT
    - **prod_price** of type REAL
    - **prod_qty** of type INTEGER
- A **primary key** is a unique value that provides a unique identifier for each record. A primary key is typically a sequence of whole numbers (int).
- The **cursor.executescript(SQL)** method executes the SQL commands to create the table. The **cursor.executescript()** method can execute multiple SQL statements
- **connection.commit()** writes the data to the file.

This is the SQL code.

```
CREATE TABLE "products" (
    "prod_id"       INTEGER,
    "prod_name"     TEXT,
    "prod_price"    REAL,
    "prod_qty"      INTEGER,
    PRIMARY KEY("prod_id")
```

Assignment 1: Add prod_desc Field

This database diagram was drawn in app.diagrams.net This diagram show the final version of our products table. The field **prod_desc** was added.



Here is our final data dictionary.

Field Name	Data Type	Description
prod_id (primary key)	INTEGER	Unique product identifier
prod_name	TEXT	Product name
prod_desc	TEXT	Product description
prod_rank	INTEGER	Video game rank 1-10
prod_price	REAL	Product price
prod_qty	INTEGER	Product quantity on hand

1. In the tutorial program, below the **prod_name** field add the SQL code to create:
    a. field **prod_desc** of type TEXT.
    b. field **prod_rank** of type INTEGER

Assignment Submission

1. Attach the program files.
2. Attach screenshots showing the successful operation of the program.
3. Submit in Blackboard.